(Refer Glasserman-Sec 2.3, Seydel(4e)-Sec 2.3)

1. Recall that if $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = e^X$, then Y is said to have a lognormal distribution. Take 3 different values for each of μ and σ^2 (total 9 cases), and simulate Y . Estimate the sample mean and variance and tabulate them along with actual values computed. Plot, for at least one set of values, the sampled distribution and the actual distribution. For the sampled distribution, compute the autocorrelation of lags $1, 2, 3$ and 4.

2. Consider the multivariate normal distribution $\boldsymbol{X} \sim N_2(\boldsymbol{\mu}, \Sigma)$.

   (a) Taking $\boldsymbol{\mu} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 9 & 6a \\ 6a & 4 \end{pmatrix}$ and for each of the four values of $a = -1, -0.5, 0, 0.5, 1$, generate 10,000 sample from the distribution of $\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2(\boldsymbol{\mu}, \Sigma)$.

   (b) For each value of a, plot the 2-dimensional histogram (please see https://plotly.com/python/2D-Histogram/) based on $10,000$ simulated values of $\boldsymbol{X}$.

   (c) Also, draw the contour plots of the actual densities on the histograms.

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